

Problem-Solution Fit

Date	29 June 2026
Team ID	N/A
Project Name	Rising Waters
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] ● Residents living in flood-prone areas ● Farmers ● Disaster Management Authorities ● Local Government Officials ● Emergency Response Teams	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] ● Limited internet access in rural areas. ● Lack of smartphones for some users. ● Low awareness about flood preparedness. ● Limited technical knowledge.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Available Solutions: ● Weather forecasting websites ● Government flood warning systems ● Manual river monitoring Pros: ● Basic weather information ● Official alerts Cons: ● Alerts may arrive late ● Less accurate for local flood prediction ● Limited real-time analysis
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**2. PROBLEMS / PAINS
+ ITS FREQUENCY [PR]**

- **Unexpected floods cause loss of life and property.**
- **Lack of accurate early flood warnings.**
- **Difficulty in monitoring river water levels continuously.**
- **Problem occurs frequently during heavy rainfall and monsoon seasons.**

**9. PROBLEM ROOT /
CAUSE [RC]**

Heavy rainfall.

- **River overflow.**
- **Poor drainage systems.**
- **Climate change.**
- **Lack of real-time flood prediction.**

**7. BEHAVIOR
+ ITS INTENSITY [BE]**

- **Regularly checks weather updates.**
- **Monitors river water levels during rainy season.**
- **Responds immediately to emergency alerts.**
- **Shares warning messages with family and neighbors.**

**3. TRIGGERS TO ACT
[TR]**

Heavy rainfall forecast.

Rising river water levels.

Weather department alerts.

Sudden climate changes.

**10. YOUR SOLUTION
[SL]**

Develop an AI & Machine Learning-based Rising Waters Flood Prediction System that:

- **Collects real-time weather and river water-level data.**
- **Predicts flood risk using AI/ML algorithms.**
- **Classifies flood risk as Low, Medium, or High.**
- **Sends instant alerts through SMS and mobile applications.**
- **Helps residents and authorities take preventive actions and reduce flood damage.**

**8. CHANNELS OF
BEHAVIOR [CH]**

ONLINE

- **Mobile Application**
- **SMS Alerts**
- **Government Websites**
- **Weather Applications**
- **Social Media**
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**4. EMOTIONS
BEFORE / AFTER [EM]**

Before:

- **Fear**
- **Anxiety**
- **Confusion**

After:

OFFLINE

- **Local Government Announcements**
- **Community Volunteers**
- **Emergency Sirens**
- **Radio & Television**
- **Disaster Management Teams**

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| <ul style="list-style-type: none">• Relief• Confidence• Better Preparedness | | |
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